

Customer Behavior Analysis

Alfido Tech Internship · Task 1

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RFM Segmentation · Purchase Pattern Analysis · Churn Risk Profiling · Dataset:
ecommerce_customer_data_large.csv · Submitted May 2026

1.Executive Summary

1 Scope of Analysis

- Comprehensive Customer Behavior Analysis on 250,000 transactions from 49,661 unique customers spanning January 2020 through September 2023

2. RFM Segmentation Applied

- Customers were classified into four segments — Champions, Loyal, At Risk, and Lost — using scored quintile-based RFM modeling.

3. Key Findings

- A 20% churn rate was identified; all four product
- categories contribute equally at ~25% each;
- Credit Card and PayPal are the top payment
- methods at 33.4% each,

4. Champion Customers

- 13,928 Champion customers were identified, representing the highest-value segment with an average lifetime value of \$21,343.

Summary Statistics

Metric	Product Price(4)	quantity	Total Purchase(4)	Customer Age
Mean	\$254.74	3.0	\$2,725.39	43
Median	\$255.00	3.0	\$2,724.00	44
Mode	\$255.00	3.0	—	—

Top Product Category: Electronics

Most frequently purchased category by transaction count 62,630 transactions across all records.

Top Payment Method: Credit Card

Credit Card accounts for 33.4% of all transactions the dominant payment mode

Gender Split

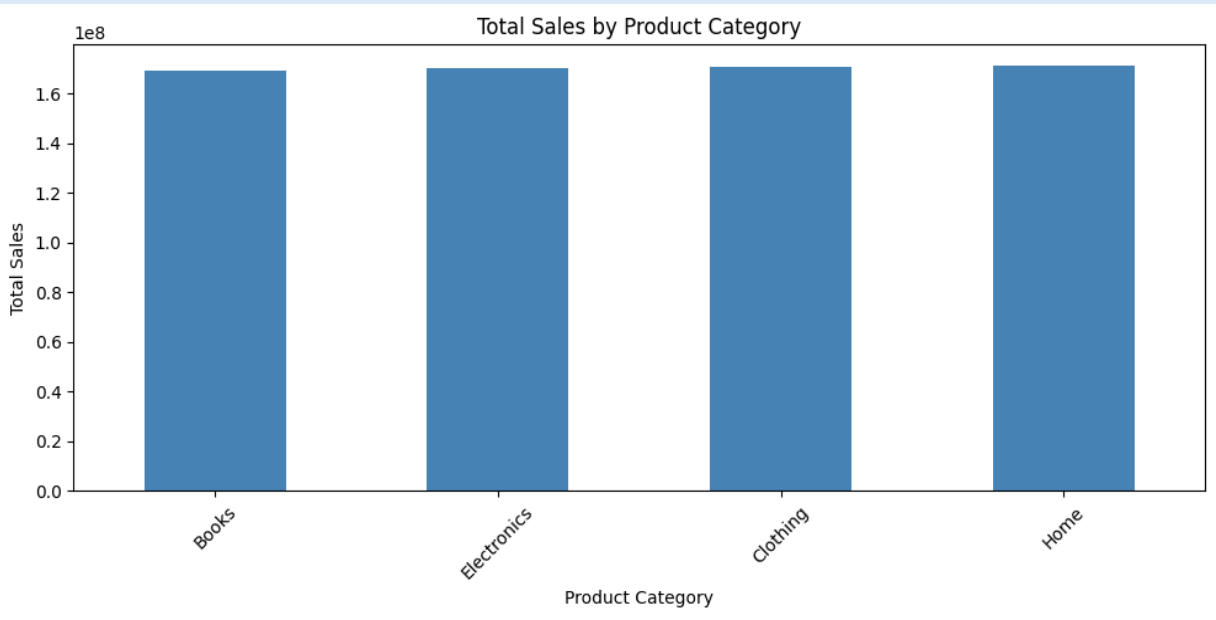
Male 50.3% Near-perfect gender balance — 50.3% Male vs 49.7% Female — with equal average spend of ~\$2,725

Core Buying Age: 35–55 Years

Core Buying Age: 35–55 Years Mean age 43.8 and median 44 confirm a mature, mid-career customer demographic.

Data Visualizations

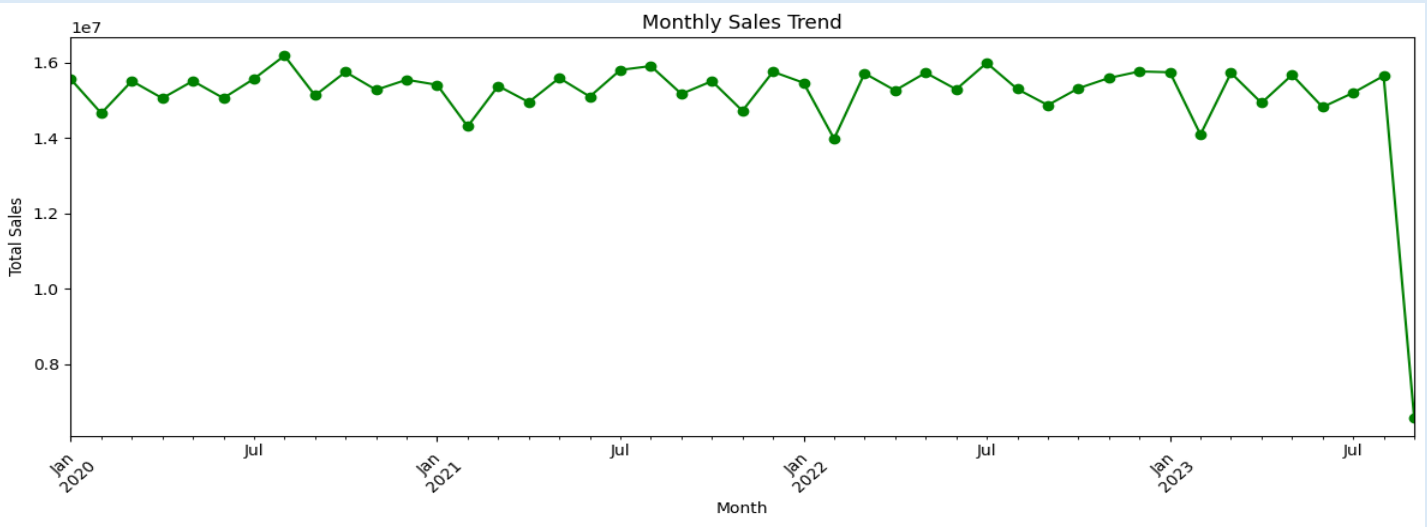
- Total Sales by Product Sub-Category



Well-Balanced Product Portfolio

All four categories contribute equally at ~25% each. Home leads at \$171.1M (25.1%), indicating no single category dominates and a diversified revenue base.

Monthly Sales Trend

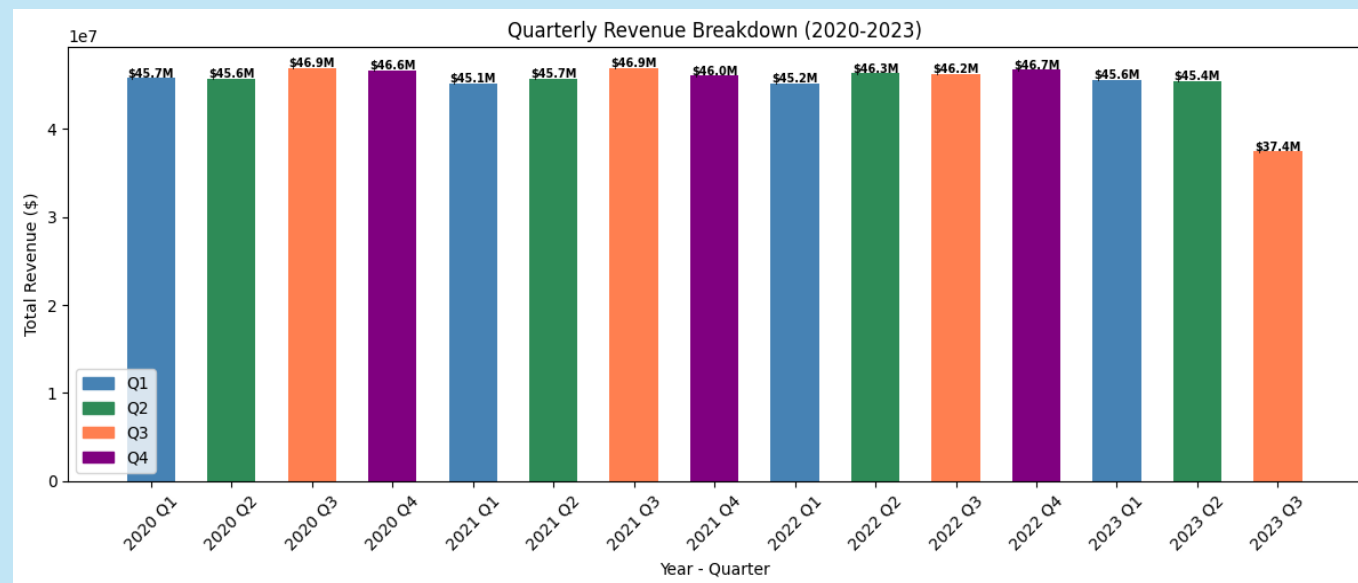


Consistent Revenue Performance

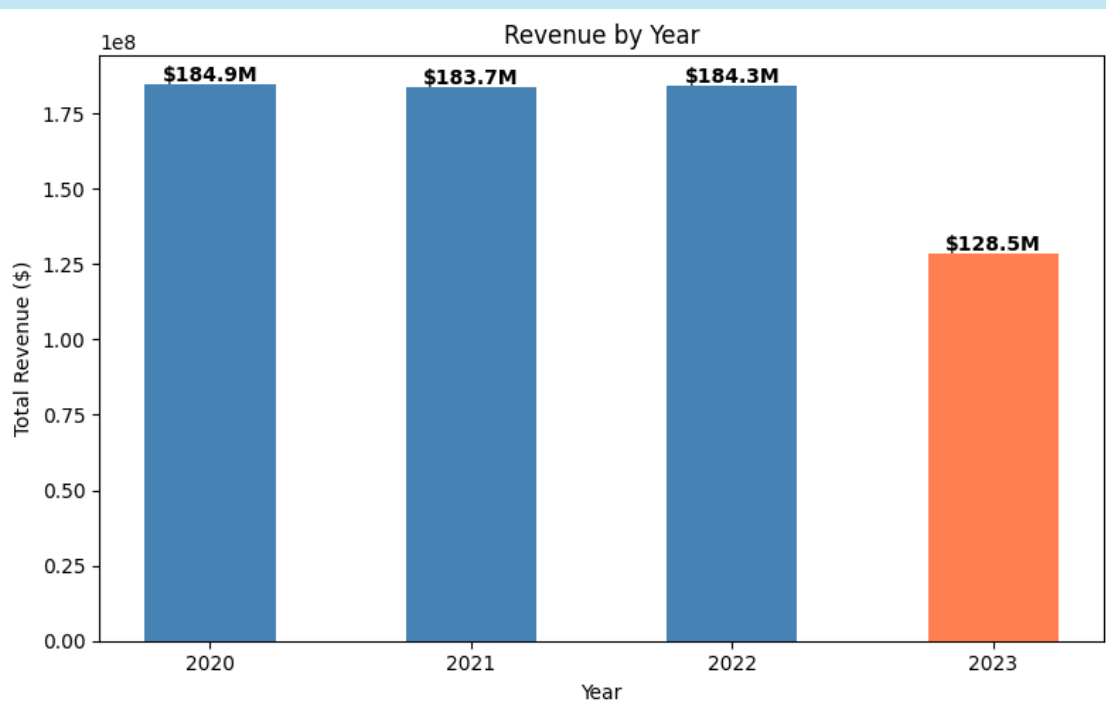
Consistent Revenue through
And remained stable at ~\$184M annually
from 2020–2022,

Quarterly Revenue

- **Stable Quarterly Performance**
Revenue remains consistent at ~\$45-47M across all quarters with no dominant seasonal spike indicating year-round stable purchasing behaviour.



Revenue by Year



- **Revenue stable**
At ~\$184M (2020-2022), 2023 lower at \$128.5M due to incomplete data.

Payment Method Distribution

➤ Credit Card Dominates

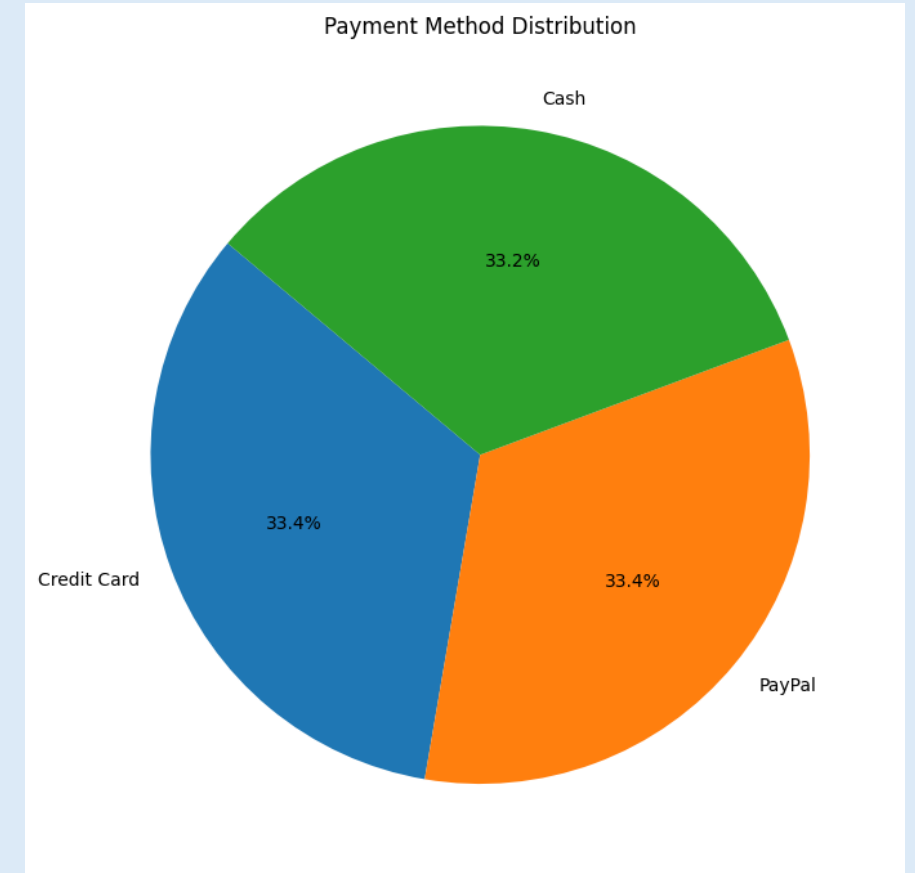
83,547 transactions (33.4%) — the clear preferred payment method across all customer segments.

➤ PayPal is Second

83,441 transactions (33.4%) — strong digital wallet adoption among the customer.

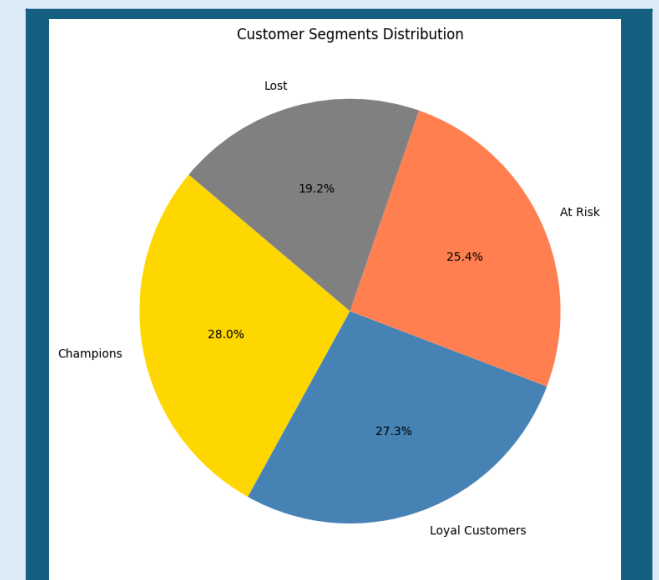
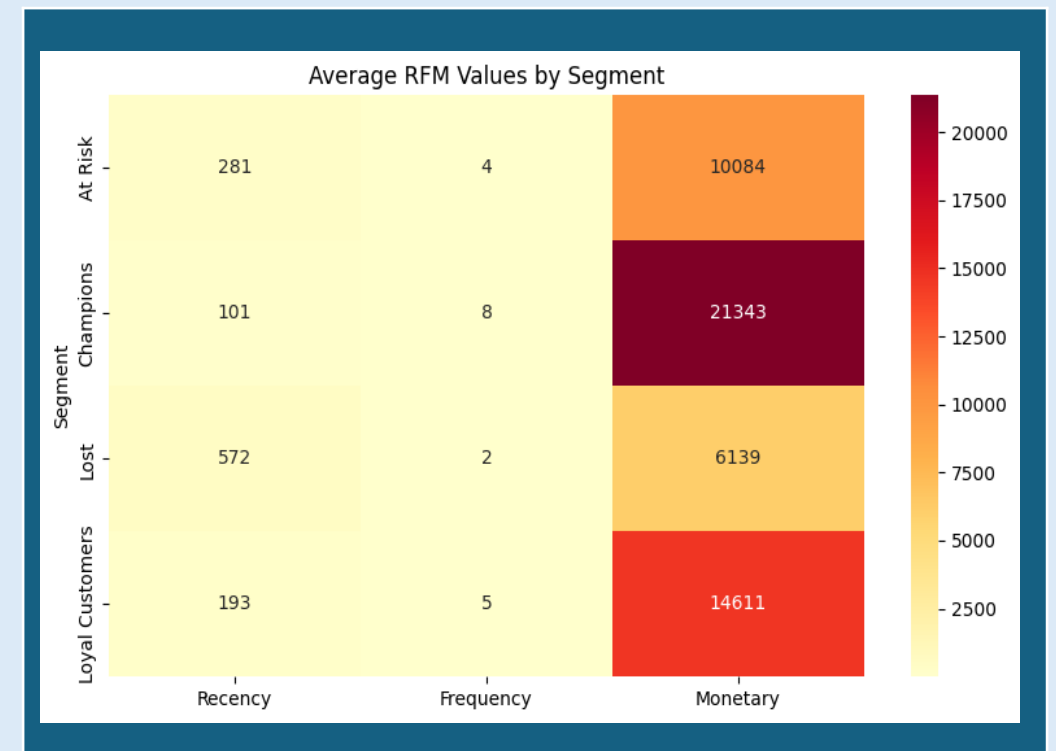
➤ Cash Still Relevant

83,012 transactions (33.2%) — indicates a portion of customers prefer traditional payment.

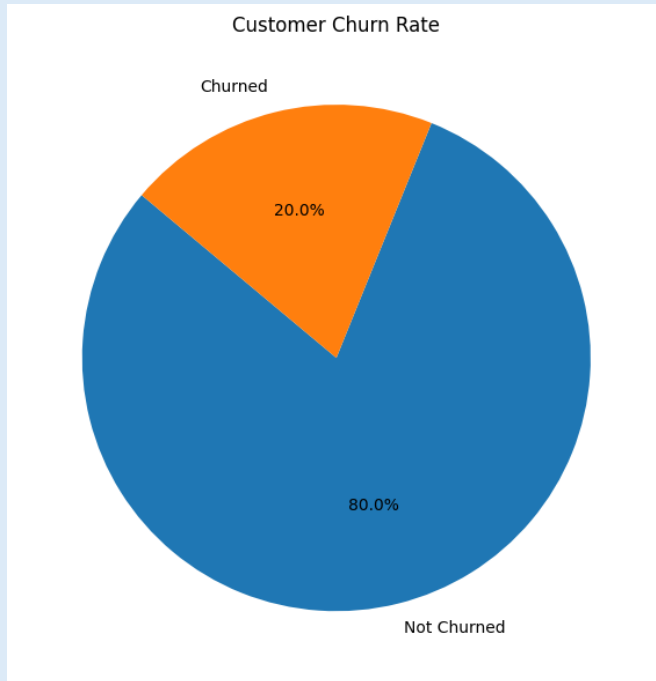


Feature Engineering- RFM Model

Segment	Score	Customers	Base %	Avg Recency	Avg Frequency	Avg Monetary
Champions	12-15	13,928	28.1 %	101 days	8txns	\$21,343
Loyal Customers	9-11	13,559	27.1 %	193 days	5txns	\$14,611
At Risk	6-8	12,637	25.5 %	281 days	4 txns	\$10,084
Lost	3--5	9,537	19.2 %	572 days	2txns	\$6,139



Churn Rate



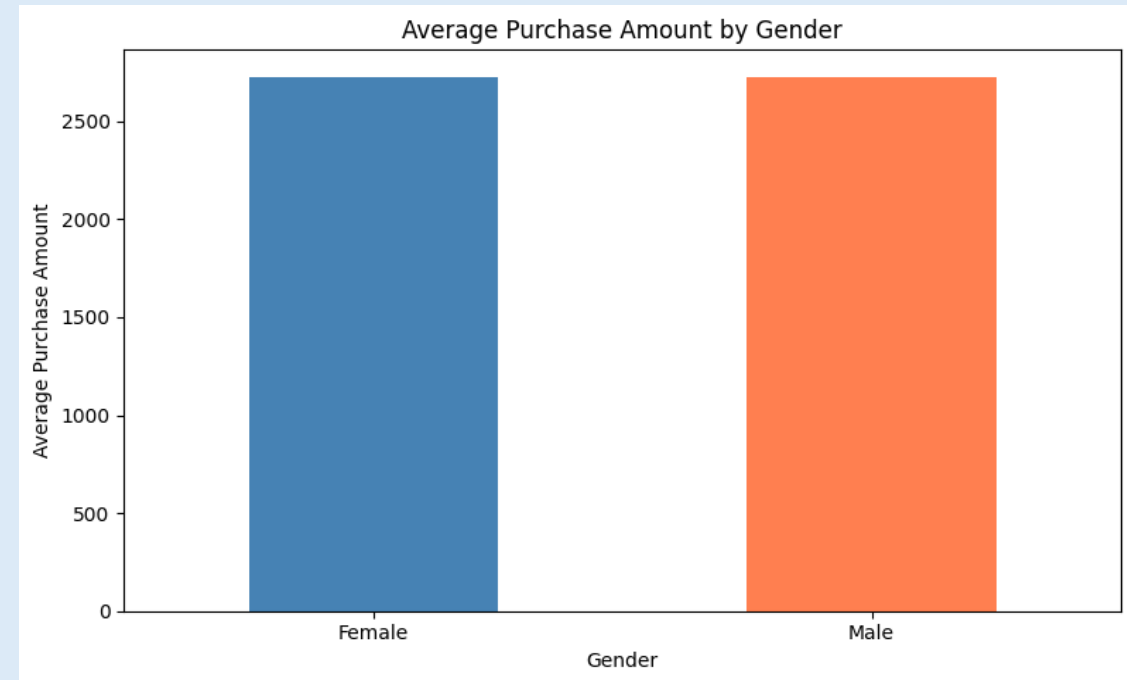
- Churn Analysis

Retained: 39,732 customers (80.0%)

Churned: 9,929 customers (20.0%)

- 1 in 5 customers has churned. The Lost RFM segment overlaps significantly with churned customers, confirming the predictive power of the RFM model.

Avg Purchase by Gender



- Gender & Age Profile

Female: 124,324 txns (49.7%)

Male: 125,676 txns (50.3%)

- Near-perfect gender balance with Male slightly leading at 50.3%.
- Equal avg spend (~\$2,725). Core buying age: 35–55 years (mean 43.8, median 44).

5.Recommendation for Alfido Tech

Launch a Champions VIP Loyalty Program

- 13,928 Champions average \$21,343 LTV. Introduce exclusive rewards ,early product access, and personalized campaigns across all four equal revenue categories (~25% each) to maximize retention.

Build a Predictive Churn Early Warning System

- With 9,929 customers lost at a 20% churn rate, train an RFM-based .Logistic Regression or Random Forest classifier to flag high-risk customers 60–90 days early and enable proactive intervention.

Deploy Win-Back Campaigns for At Risk Customers

- 12,637 At Risk customers have been inactive for 281 days. A time-sensitive 10–15% discount or free-shipping offer sent within 30 days can recover this segment before permanent loss.

Grow All Categories via Targeted Promotions

- All four categories — Books, Electronics, Clothing and Home — contribute equally at ~25% revenue each. Bundle deals, influencer partnerships and category-specific discounts can boost revenue uniformly across all product segments.

Introduce Credit Card & PayPal Incentives

- Credit Card and PayPal each dominate at 33.4% of transactions. Offer 2% cashback on both payment methods to increase transaction frequency and improve customer satisfaction across both dominant payment segments.

Conclusion

Alfido Tech has a strong Champion & Loyal base (55.2% combined) but faces an urgent 20% churn challenge and a disengaging At Risk segment (25.5%). Immediate focus on retention, personalization, and payment incentives across Credit Card and PayPal will be key to sustainable growth. All code and visualizations are available in the linked Jupyter notebook. Dataset and report uploaded to GitHub for submission.

Strong Foundation

Strong Foundation
Champions (28.1%) and Loyal Customers (27.1%) combined represent 55.2% of the base — a powerful retention asset ready for a VIP loyalty progress

Urgent Priority

20% churn rate (9,929 customers) and a disengaging At Risk segment (25.5%) demand immediate win-back campaigns and predictive RFM-based modeling.

Growth Opportunity

All four categories contribute equally at ~25% each. Credit Card and PayPal incentives at 33.4% each can unlock significant incremental revenue growth.